



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECD 2613
SYSTEM ANALYSIS AND DESIGN

PROJECT : PHASE 2
Information System Gathering and Requirement

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1.0 Overview of the Project

In this project, we will collect information requirements and look at system requirements for this project. This project will determine the information needs that can help you grasp the 4W1H of an operational system. To gather the information, we interviewed the client with some questions about workflow and condition of the current system. After gathering the requirements information, we have a solid understanding of the business function based on a piece of precise information about the people, objectives, data, and procedures involved.

The business flow of the AS-IS system for task management is then visualised using context diagrams in the following steps which are described as a high-level Data Flow Diagram (DFD), a context diagram depicts the information flow between the taskmanagement.com and external entities, including users and lecturers.

Data flow diagrams, which are the parent and child diagrams, are used to depict the external entities, data flows, data stores, and procedures involved once the context diagram has been created. Parent diagram is known as a level 0 diagram which is the explosion of context diagram. An integer is assigned to each process. The parent diagram includes all external entities as well as the main data storage. Each process in the parent diagram will be exploded to generate a child diagram. A child diagram utilises the same number as the parent diagram, receives the same input, and produces the same output.

When the tasks are finally completed, we now thoroughly understand the current system and data flow diagrams, entity relationship diagrams, and information requirements.

2.0 Problem Statement

2.1 Fragmented Task Management

Postgraduates struggle with fragmented task management processes, such as manually organising research materials, coordinating with co-authors, and tracking deadlines for paper submission.

2.2 Communication Gaps

Communication breakdowns between students and supervisors, as well as among co-authors, result in misunderstandings, missed deadlines, and delays in the publication process.

2.3 Resource Allocation Issues

Difficulty in allocating resources effectively, such as time, research materials, and collaboration tools, leads to inefficiencies and suboptimal outcomes in research and publication endeavors.

2.4 Deadline Management

Poor management of deadlines for literature review, data collection, analysis, and writing stages hampers progress and increases the likelihood of rushed, incomplete submissions.

2.5 Collaboration Challenges

Coordinating tasks and sharing progress among co-authors or research team members becomes cumbersome due to reliance on disparate communication channels and tools.

2.6 Publication Requirement Compliance

Lack of awareness or reminders regarding publication requirements, formatting guidelines, and submission procedures results in unnecessary revisions and rejections from journals or conferences

3.0 Proposed Solutions

Task Management System (TMS) is a system that allows postgraduate students to manage their assignments and projects efficiently. The system can be installed and applied on mobile devices such as mobile phones and tablets in order to enhance the functions of the Task Management System (TMS).

The Task Management System's (TMS) first function assists users or postgraduate students in allocating assignments so they can schedule a time to finish the projects. Apart from that, users can simply adhere to the timelines while working on projects because the system automatically saves the work timelines into a spreadsheet file each time a user opens and closes a project file. Additionally, the users are able to monitor the status of each work and the projects' progress. The users are required to enter the task progression from 0% to 100%. As a result, users can determine if the task is finished or only halfway done.

The second feature is the specialized collaboration and communication tools. The co-author of a project can be added by the author or postgraduate students who collaborate on the project. While the partners are now offline, users can still discuss projects with them by leaving a message in the communication space. As soon as they get the notification, the partners can reply to the message.

The third feature is the notification feature. The system will notify the users when they received messages, the upcoming due date, the existence of overdue tasks. The system helps users manage their time so they can complete their project publication and keeps the projects on schedule.

Besides that, the Task Management System uses a mobile platform, it requires a mobile gadget and Internet access to use the system. A server and database system are required for this system since the system has to store user projects' progression and communicate with the team members. A security system guards the system to prevent confidential project information from being discovered by unidentified third parties.

Furthermore, the Task Management System requires permission to access dates, times, and files to track the progress of the project and manage the file. The message and project timeline are stored in a database within the system. Users can search the message history for messages sent by other team members. All the information and data are password-protected, the data can be accessed by the users with the password only.

4.0 Information Gathering Process

The information gathering process is one of critical processes in system analytics and design as it provides insights into user needs and requirements. The main purpose of this information gathering process is to get detailed and qualitative data from potential users and stakeholders.

4.1 Method Used

For our Task Management System project, the method that was chosen as our primary data collection method is interviewing. By using this method, we have collected data on human and system information requirements. This interviewing session is conducted online with Mr. Muhammad Aliif Ahmad through the application “Webex”. We have used open-ended questions and closed questions regarding six major aspects which are background, current manual process, desired features and functionalities, technical and operational constraints, stakeholder need & expectation and costs and budgets.

4.1.1 Interviewing

Here are some questions that we had asked during the interviewing session:

Open-ended questions :

1. What goals do you aim to achieve with the introduction of this system?
For this question, Mr Aliif, who is one of the stakeholders, expects to benefit from a more simplified way to manage their students. The goal is to create a unified system that will streamline the management process, track progress, and provide reminders using different types of apps and workflows.
2. How will better suggestions be collected and incorporated into the system development?
Mr. Aliif said that there is one option which is to include a feature within the system where users can suggest improvements. These suggestions can be upvoted by other users, and those with the most support can be considered for implementation in future updates.
3. What factors do you consider when selecting technologies or platforms for a new system?
We have received the answer from Mr. Aliif stating that the main factors are cost-effectiveness and adaptability to multiple operating systems. For example, using Laravel for development because it can be easily packaged for different operating systems like IOS and Android.
4. What are the specific challenges with the current manual process that led to the idea of developing a new task management system?
The main challenges faced by Mr. Aliif and his team include inefficiencies in tracking progress, managing multiple versions of documents, and the need for reminders. The

current manual process is fragmented across various tools like WhatsApp and email, which leads to lost communications and missed deadlines.

5. What is your vision for this system in terms of its use by supervisors and students?

Mr. Aliif told us that his vision for this system is to become the default tool used by all supervisors and students for managing tasks related to paper writing, publication, and overall communication. The system should simplify the process and make it more efficient.

6. How do you plan to handle the training and operational aspects of the system once it is developed?

The answer of Mr. Aliif is training is essential and should include comprehensive documentation of all functions. After development, a dedicated person will be hired to manage and maintain the system, ensuring it runs smoothly and any issues are promptly addressed.

Closed questions :

1. Are there specific metrics or key performance indicators that you hope to improve with this system?

Mr. Aliif responds that the key performance indicators for the system include the ability for students to create tasks, supervisors to check tasks, scheduling and tracking progress, and facilitating communication between students and supervisors. The goal is to replace manual processes with automated reminders and notifications.

2. Are there any specific integrations with other tools or software that would be essential to this new system?

Mr. Aliif said that for the specific integrations is to develop in-house features to manage collaboration and communication which ensures that all functionalities are within one unified system.

3. What are the budgetary limitations for developing and implementing the new system?

Mr. Aliif and his team have a limited budget. Thus, they use open-source tools like Laravel for development and Figma for high-fidelity prototypes. Whereas, the budget will cover servers, hosting, and a dedicated person for maintaining the system post-development.

4. What technical and operational constraints do you anticipate facing during the development and implementation of the new system and how do you plan to address them?

Technical constraints including managing servers and hosting were faced by them during the development and implementation of the new system. Operational constraints involve hiring someone to maintain the system after development. Mr. Aliif and his team plan to use open-source tools and ensure budget allocation for ongoing maintenance and hosting.

4.2 Summary from method used

Through this interviewing method, sharing of Mr. Muhammad Aliif Ahmad has provided valuable insights into the needs and pain points of potential users. This information will guide the design and development of the Task Management System, ensuring it meets user requirements, addresses their challenges and reduces potential problems that our team may meet.

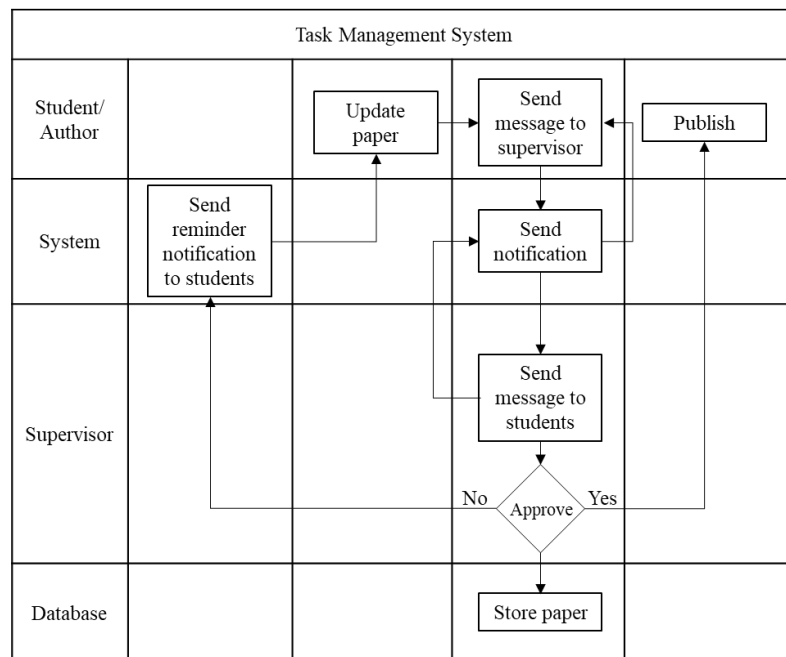
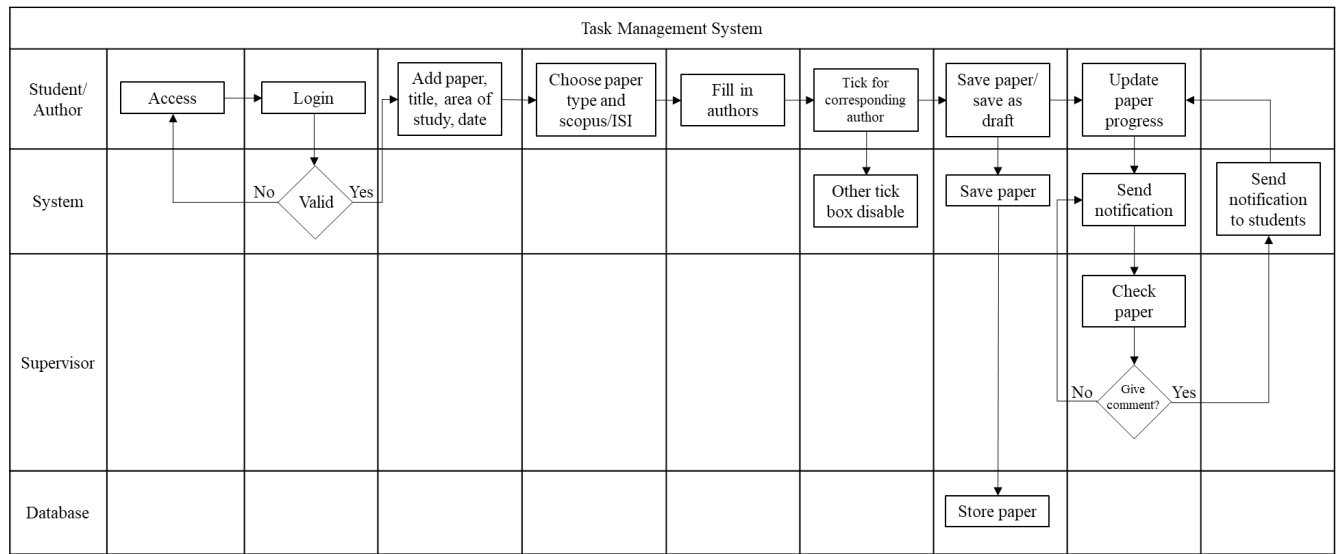
5.0 Requirement Analysis (Based on AS-IS analysis)

5.1 Current business process (scenarios, workflow)

Below are the scenarios and workflow on how the task management system current business process:

1. Users access their account using their university email.
2. Users log in to the system.
 - 2.1 If users fail to login, users are required to attempt login again.
3. Students can add paper and fill in their title, area of study and choose the date from the calendar.
4. Choose the paper type and scopus/ISI.
5. Fill in the first author and co-author up to 13 people (if required).
6. Students need to tick for the corresponding author in the tick box.
 - 6.1 Once there is a tick box that has been ticked, the other tick box will be disabled.
7. Then students can choose to save the paper or save it as a draft.
8. After students have added paper, they can update their paper progress so that they won't miss out any phases.
 - 8.1 Once the students update the progress, there will be a notification sent to their supervisor and also other authors so that everyone can always follow up the progress/changes of the paper.
 - 8.2 There will be a calendar in the dashboard that records the progress and deadlines of the paper for all dates and times so that students can refer easily.
9. In this system, there will also be a reminder feature. When the deadline is near, the system will automatically send notification to the students to notify them to update their paper on time.
 - 9.1 Supervisor will also be notified to check student's papers if the supervisor did not give any comment 3 days after the student's update.
10. Students can always message their supervisor for any question in this system so that their messages will not be obscured by other messages.
 - 10.1 Students and supervisors will receive the notification of the messages immediately so that they won't miss out on any messages.
11. Once the students have done the paper and get approval from the supervisor, they can prepare to publish their paper.

Workflow:



5.2 Functional Requirement (input, process and output)

process	input	output
Add Paper	Paper Title, Area of Study, Paper Type (Journal, Conference, Book Chapter), Indexing (ISI, Scopus), Date (with calendar), First Author, Co-authors (optional), Check if Complete, Save Paper, Draft	Paper Added Confirmation
Track Paper Progress	Paper Details	Progress Updated Confirmation
Update Paper Progress	Paper Details, Progress Status	Progress Updated Confirmation
Notify Supervisor	Paper Details, Progress Status	Notification Sent to Supervisor
Set Deadlines	Paper Details, Submission Dates	Deadline Notifications
Connect with Deadlines	Deadlines (extracted from paper submissions), Calendar Updates	Calendar with Deadlines
Notify on Deadlines	Deadlines	Notifications for deadlines
List Authors based on Paper	Paper Details, Author Details	Author List
Update author list	Paper Details, Author Details	Author List Updated Confirmation
Generate Paper Statistics	Paper Details, Author Details, Submission Dates, Indexing Information	Report with Statistics
Create Statistics	Report Details	Generated Report

5.2.1 Context Diagram

system	input	output	Entities
Paper Management System	User Credentials, Paper Details, Deadlines, Approval, Collaborator Information	Access Confirmation, Notifications, Reports	Users, Supervisors, Collaborator

5.2.2 Level 0 Diagram

process	input	output
Login	User Login Details, User Detail	User Login Details, Login Confirmation
Add and Manage Papers	Paper Details, Collaborator Information	Paper Added Confirmation, Paper Details, Collaborator Information
Track and Update Progress	Paper Details, Progress Updates, Comments	Paper Detail, Progress Updated Confirmation, Progress Updates, Notification
Manage Calendar and Deadlines	Deadline Dates	Calendar Updates, Deadline Notification
Generate Reports	Paper Details, Author Details, Completed Paper, Approval	Completed Paper, Paper Detail, Generated Reports

5.2.3 Level 1 Diagram

5.2.3.1 Login

process	input	output
Check User Detail	User Login Detail	User Detail
Check User Password	Password	Login Confirmation

5.2.3.2 Add and Manage Papers

process	input	output
Select Paper Type	Paper Details	Paper Details
Enter Paper Title	Paper Details	Paper Details
Enter Area of Study	Paper Details	Paper Details
Enter Collaborator	Paper Detail, Collaborator Information	Paper Detail, Collaborator Information
Save Paper	Save Paper Button	Paper Detail, Collaborator Information, Paper Added Confirmation

5.2.3.3 Track and Update Progress

process	input	output
Update Progress	Paper Details, Progress Status	Progress Updated Confirmation, Progress Status
Notify Supervisor	Paper Details, Progress Status	Notification

5.2.3.4 Manage Calendar and Deadlines

process	input	output
Connect Deadlines to Calendar	Deadline Dates	Deadlines
Notify Users on Deadlines	Calendar Integration	Notifications

5.2.3.5 Generate Reports

Process	Input	Output
Submit Complete Paper	Completed Papers	Paper Details, Completed Papers
Authorisation by supervisor	Paper Details	Approval
Compile Paper Statistics	Paper Submission Details, Author Details, Approval	Report Details
Display Report	Report Details	Generated Report

5.3 Non-functional Requirement (performance and control)

1. Performance

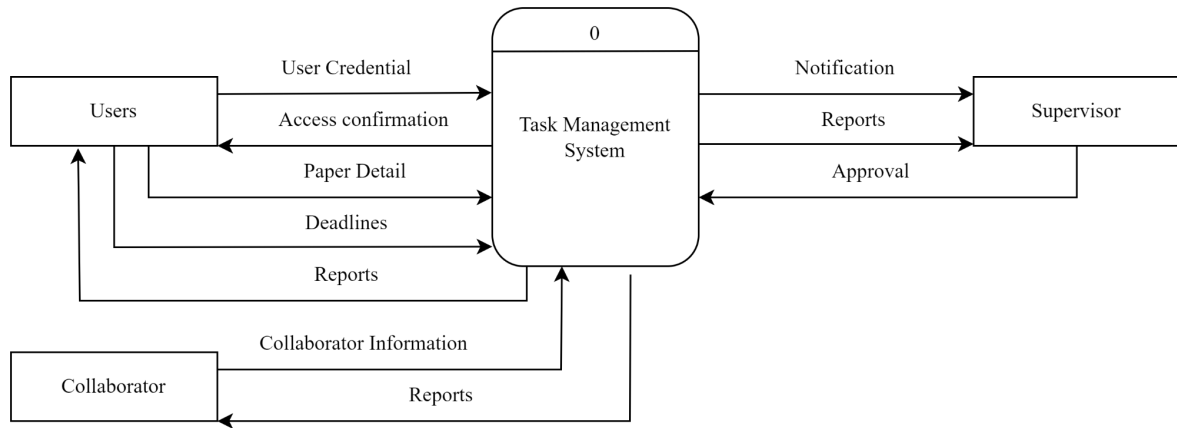
- a. The Task Management System (TMS) must be available 99.9% of the time, ensuring minimal downtime for users.
- b. The system should have a response time of less than 2 seconds for most user actions, such as logging in, submitting proposals, and receiving feedback.
- c. The system must be scalable to support a growing number of users, including students, supervisors, and administrators, without performance degradation.
- d. The system must process and update task status, notifications, and user actions in real-time to provide immediate feedback and updates.

2. Control

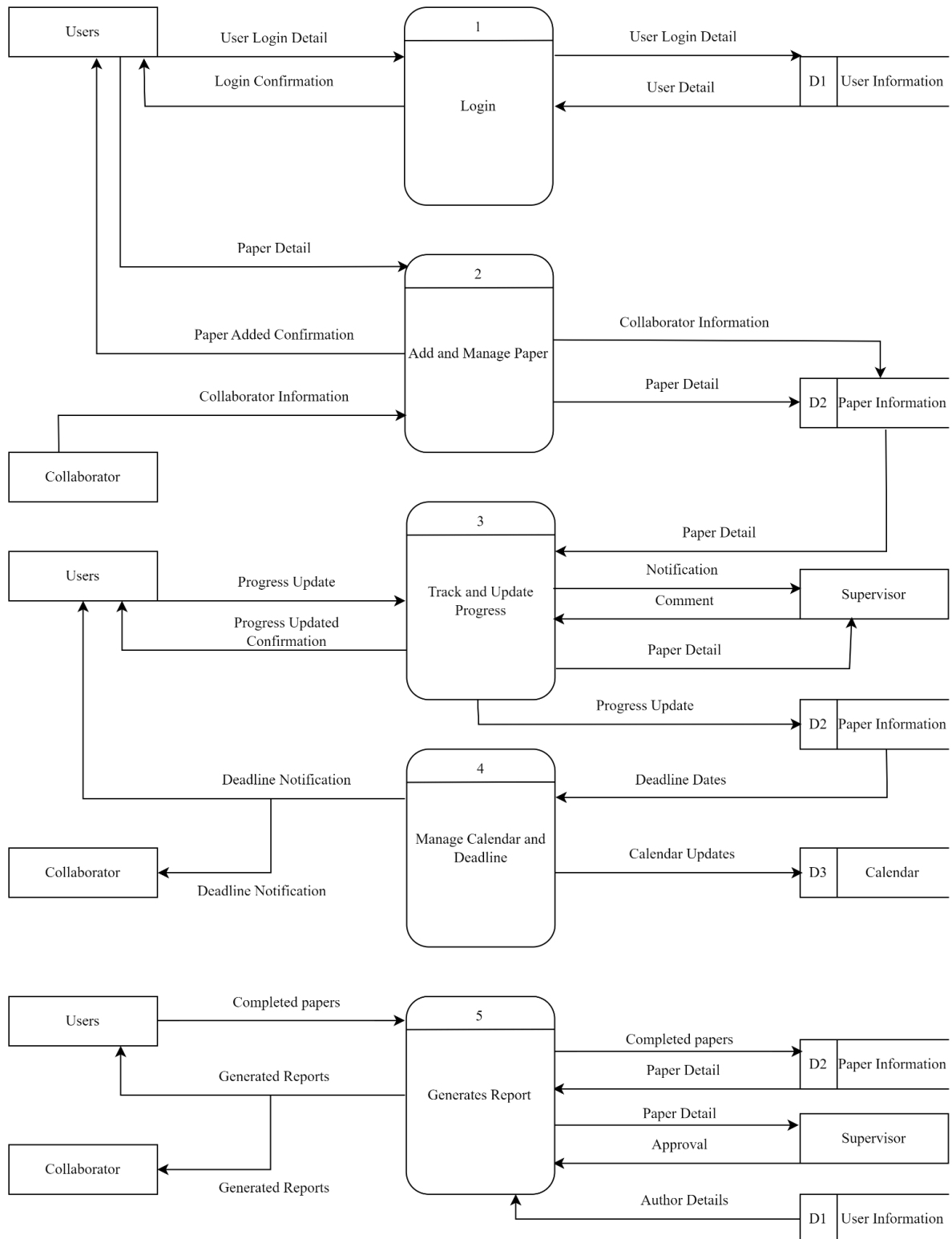
- a. The system must provide secure login mechanisms, using university email accounts and strong password policies.
- b. The system should provide comprehensive user support, including a helpdesk, FAQs, and user guides, to assist users in resolving issues quickly and efficiently.
- c. The system must comply with relevant regulations and standards, such as GDPR for data protection and FERPA for students privacy, to ensure legal and regulatory adherence.

5.4 Logical DFD AS-IS system (Context Diagram, Level 0 Diagram, Child)

5.4.1 Context Diagram

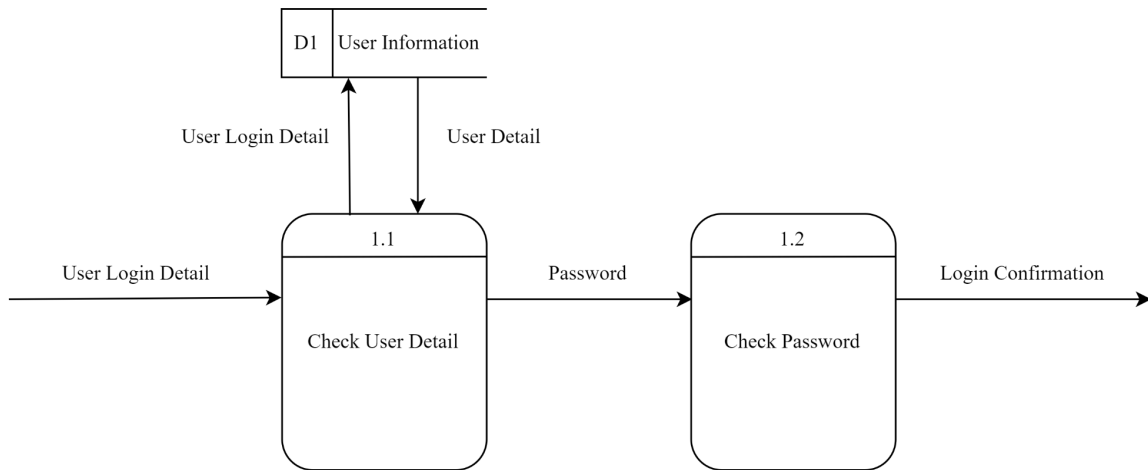


5.4.2 Level 0 Diagram

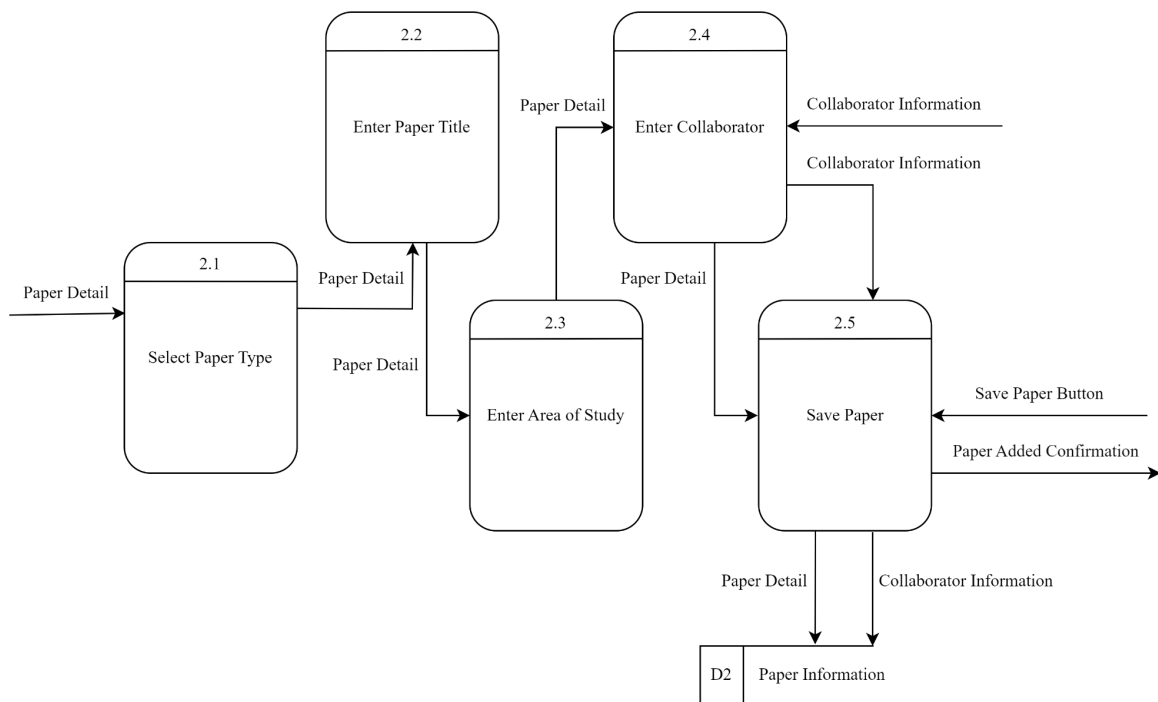


5.4.3 Child Diagram

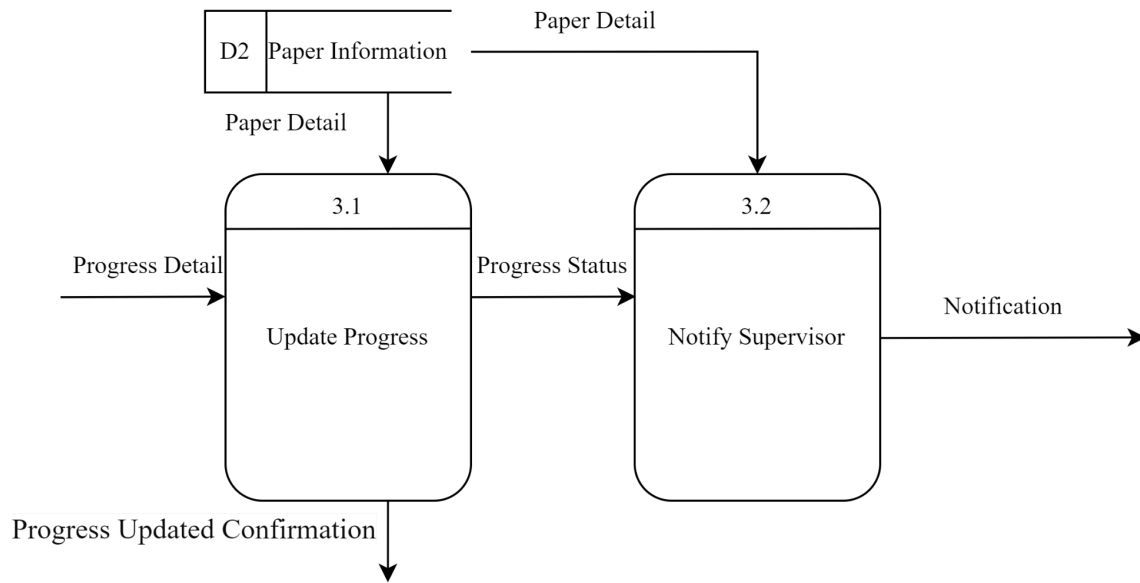
5.4.3.1 Login



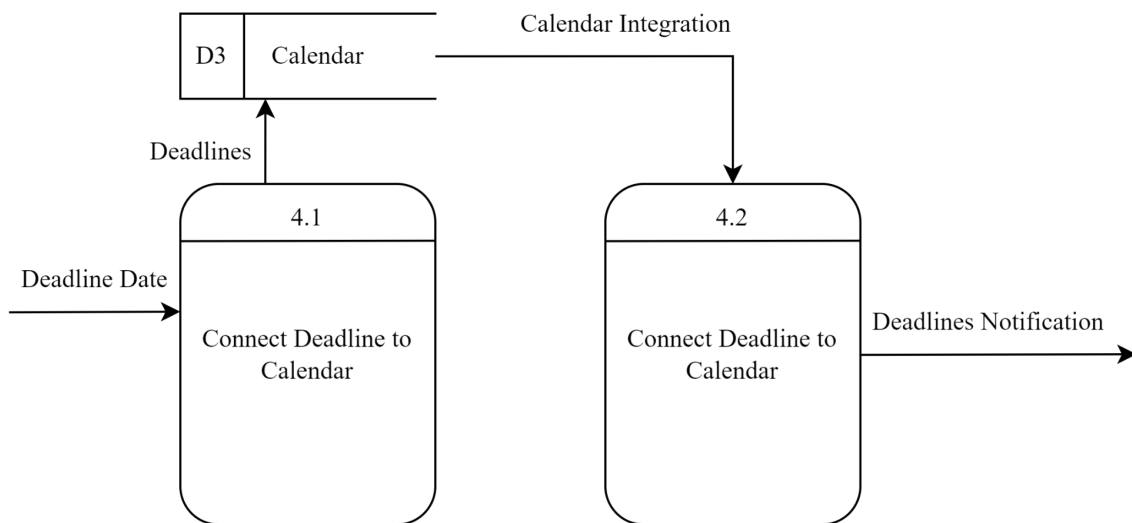
5.4.3.2 Add and Manage Paper



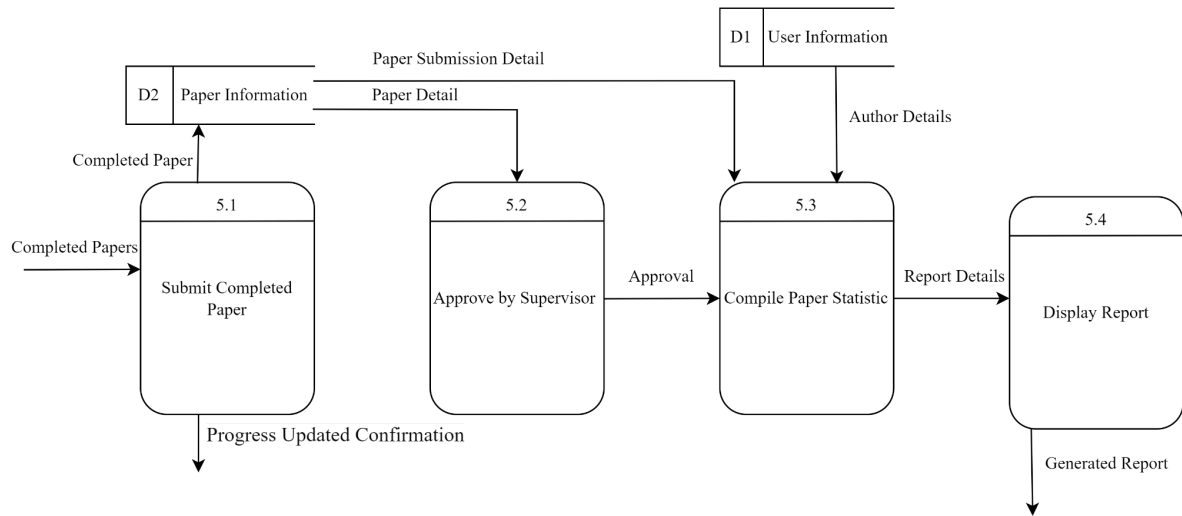
5.4.3.3 Track and Update Progress



5.4.3.4 Manage Calendar and Deadlines



5.4.3.5 Generate Reports



6.0 Summary of Requirement Analysis process

To sum up, after gathering required information via interview and doing the requirement analysis process, we notice that the As-Is “Task Management” system has several parts that can be improved so that it can be more efficient. Moreover, some features can also be added to the system to decrease the workload.

The following are the new functions and features that would be added in the To-Be system:

1. Supervisor checking system
2. Improved authors' details database
3. Calendar management
4. Plagiarism checker
5. Database for paper details
6. Paper writing tracking process